Practical No. 10

IDS Evasion through Stealth Scanning

Tool: Nmap

Theory

Nmap is an open-source network scanning tool used to discover hosts, open ports, running services, and operating systems on a network. Intrusion Detection Systems (IDS) monitor network traffic for suspicious scanning activity. Nmap provides stealth scanning techniques such as SYN scan, FIN scan, and NULL scan that are designed to avoid triggering IDS alerts. This experiment is performed only on Metasploitable 2 in an isolated lab network.

Steps

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1. Start Kali Linux and Metasploitable 2 on a Host-Only network in VirtualBox. Verify Nmap is installed. Run: nmap --version

2. Perform a basic ping scan to discover live hosts on the network. Run: nmap -sn 192.168.56.0/24 — This lists all active hosts without scanning ports.

3. Run a standard TCP connect scan on the target to find open ports. Run: nmap -sT 192.168.56.102 — This completes the full TCP handshake and is easily detected by an IDS.

4. Run a SYN stealth scan which sends only SYN packets and does not complete the handshake, making it harder to detect. Run: sudo nmap -sS 192.168.56.102 — Compare the results with the previous scan.

5. Run a FIN scan which sends FIN packets instead of SYN. Many IDS systems do not flag FIN packets as a scan. Run: sudo nmap -sF 192.168.56.102

6. Run a NULL scan which sends packets with no flags set. Run: sudo nmap -sN 192.168.56.102 — Open ports typically do not respond while closed ports send a RST packet.

7. Perform service and version detection to identify what software is running on each open port. Run: nmap -sV 192.168.56.102

8. Run an OS detection scan to identify the target operating system. Run: sudo nmap -O 192.168.56.102

9. Save the full scan results to a file for documentation. Run: nmap -A 192.168.56.102 -oN scan_results.txt — View the file. Run: cat scan_results.txt

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Conclusion

Nmap successfully performed standard and stealth scans on Metasploitable 2, identifying open ports, running services, and the target OS. The stealth scans such as SYN, FIN, and NULL demonstrated techniques used to reduce detection by IDS systems during network reconnaissance.